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TECHVISION CONSULTING INC.
Emerging Technologies Overview and Its Areas

A Comprehensive Report

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1. Executive Summary

This report presents a comprehensive analysis of emerging technologies, prepared to guide the client recommendations and strategic planning work of TechVision Consulting Inc. All the information is acquired from the Module 1 lecture of DCIT 26 - Application Development and Emerging Technologies, which examines how technologies emerge in the real world and how they can be applied across a variety of domains.

The report first establishes what qualifies a technology as "emerging," then examines the five defining characteristics of emerging technologies identified by Rotolo, Hicks, and Martin (2015): radical novelty, relatively fast growth, coherence, prominent impact, and uncertainty and ambiguity. It then profiles the major areas of emerging technology highlighted by the World Economic Forum: Augmented Reality (AR), Artificial Intelligence (AI), Machine Learning, and Quantum Computing; before situating these within the Fourth Industrial Revolution (Industry 4.0), including its four historical stages and its core building-block technologies: Big Data Analytics, the Internet of Things (IoT), Cloud Computing, and Augmented Reality.

2. Introduction and Purpose of the Report

As a Technology Analyst at TechVision Consulting Inc., a firm that advises businesses and government agencies on technology adoption, I prepared this report in response to the management's request for a comprehensive briefing on emerging technologies. The goal is to provide an evidence-based foundation for advising clients on which technologies to monitor, how to understand their maturity, and what strategic approach to take toward them.

The scope of this report is limited strictly to the concepts presented in the Module 1 lecture. It is divided into four parts: (a) the definition and meaning of emerging technology; (b) the five characteristics that distinguish an emerging technology; (c) the principal areas of emerging technology identified by the World Economic Forum; and (d) the Fourth Industrial Revolution as the historical and industrial context for the development of these technologies.

3. Understanding Emerging Technologies

Emerging technology means different things to different people, and the lecture notes that considering only new technologies as "emerging" does not capture the full picture. A few people view emerging technologies as advancements of existing innovations that are applied creatively to achieve modernity. Under this view, a technology can remain "emerging" for decades if it continues to progress and find new applications.

Technology can also be identified as emerging through its characteristics rather than through definitions alone. The following section examines the five characteristics identified by Rotolo, Hicks, and Martin (2015).

4. The Five Characteristics of an Emerging Technology

According to Rotolo, Hicks, and Martin (2015), emerging technologies can be recognized through five main characteristics. The table below summarizes each characteristic.

Characteristic	Core Description
Radical novelty	Emerging technology is radically new, or existing technology is reused to perform a task using a different basic principle than previous technology. For example, artificial intelligence applications are used to achieve a variety of results.
Relatively fast growth	Emerging technologies show faster growth rates than non-emerging technologies in terms of actors involved, funding, knowledge outputs, prototypes, products, and services. For example, the number of users of artificial intelligence is increasing across various companies, universities, and individuals, indicating a visible expansion in every dimension of growth.
Coherence	Emerging technology represents a convergence of previously separated research streams and technologies that are maintained over time due to internal logical interconnections. One example is the grouping of cloud computing, artificial intelligence, data analytics, and robotics to achieve various goals: previously separate fields are now logically interconnected and moving together.
Prominent impact	Emerging technology benefits multiple sectors, transforms industries, and has a significant economic impact across domains. Artificial intelligence shows this: it has a significant impact and causes big changes in a variety of sectors, including business, academic, healthcare, and individuals around the world.
Uncertainty and ambiguity	The impact of emerging technology lies in the future; applications remain malleable, outcomes are incompletely known, and different social groups attach different meanings to technology. For example, some people can predict a variety of artificial intelligence applications or uses, but these applications may or may not be developed successfully.

In conclusion, these characteristics define an emerging technology as radically novel and relatively fast-growing, with a degree of coherence that persists over time. Such technologies have the potential to have a significant impact on socioeconomic domains, including actors, institutions, interaction patterns, and knowledge production processes. Their most significant impact, however, is in the future, so the emergence phase remains uncertain and ambiguous. (Rotolo, Hicks, & Martin, 2015).

5. Areas of Emerging Technology

The World Economic Forum states that the following areas of emerging technology have the potential to change the long-standing industry practices in the future.

Emerging Technology Area	Essence	Examples / Notes
Augmented Reality (AR)	Overlays computer-generated display, sound, text, and effects onto the user's real-world view. It creates a cohesive but improved view of the world by fusing real and computer-generated scenes and images.	In-car driving and reverse-parking guides; Pokémon GO; virtual furniture previews before purchase.
Artificial Intelligence (AI)	Simulation of human intelligence processes (learning, reasoning, self-correction) by machines.	Progressing since the 1950s; still emerging due to ongoing research and new applications.
Machine Learning	AI application that lets machines learn from experience like humans.	Healthcare (rapid identification and development of new medicines); financial and marketing services; transportation.
Quantum Computers	Quantum mechanics is used by computers to solve problems effectively; uses qubit as their basic unit of computation.	Demonstrated by IBM, Google, Rigetti Computing, IonQ; future potential to simulate atoms/molecules and aid materials design.

6. The Fourth Industrial Revolution as Context

6.1 The Four Stages of the Industrial Revolution

Production technology is changed by technological advancements. The stages of production technology are referred to as the "industrial revolution." The industrial revolution is the methodical transformation of manufacturing by integrating digital and physical systems to boost efficiency, reduce costs, and improve quality. As a result, production technologies significantly changed people's lifestyles and working conditions.

Stage	Period	Key Development
First Industrial Revolution (Industry 1.0)	18th century	Beginning of mechanical production using hydroelectric and steam-powered systems.
Second Industrial Revolution (Industry 2.0)	19th century	New technological systems that use electricity were introduced, allowing for increased production

		and more sophisticated processes.
Third Industrial Revolution (Industry 3.0)	First computer era	Information technology (IT) and electronics have evolved to further automate production by making use of internet access, connectivity, and renewable energy.
Fourth Industrial Revolution (Industry 4.0)	Current and developing environment	Disruptive, cutting-edge technologies (IoT, robotics, VR, AR, and AI) are changing the way we live and work, altering traditional production relationships among suppliers, producers, and customers, as well as between humans and machines.

6.2 Building-Block Technologies of Industry 4.0

Some of the technologies that form the building blocks of Industry 4.0 include the following:

- **Big Data Analytics** - The process of collecting, organizing, and evaluating massive data sets from various sources to discover patterns and other useful information.
- **Internet of Things (IoT)** – It is a computing concept that describes how everyday physical objects can be connected to the internet and communicate and interact with other devices. This could include sensor and wireless technologies.
- **Cloud Computing** - Refers to the delivery of hosted services over the Internet.
- **Augmented Reality in Industry 4.0** - AR-based systems enable a wide range of services, including part selection in a warehouse and repair instructions delivered via mobile devices.

7. Strategic Implications for TechVision Clients

Drawing exclusively on the concepts covered in the lecture, the following recommendations guide our client advisory work:

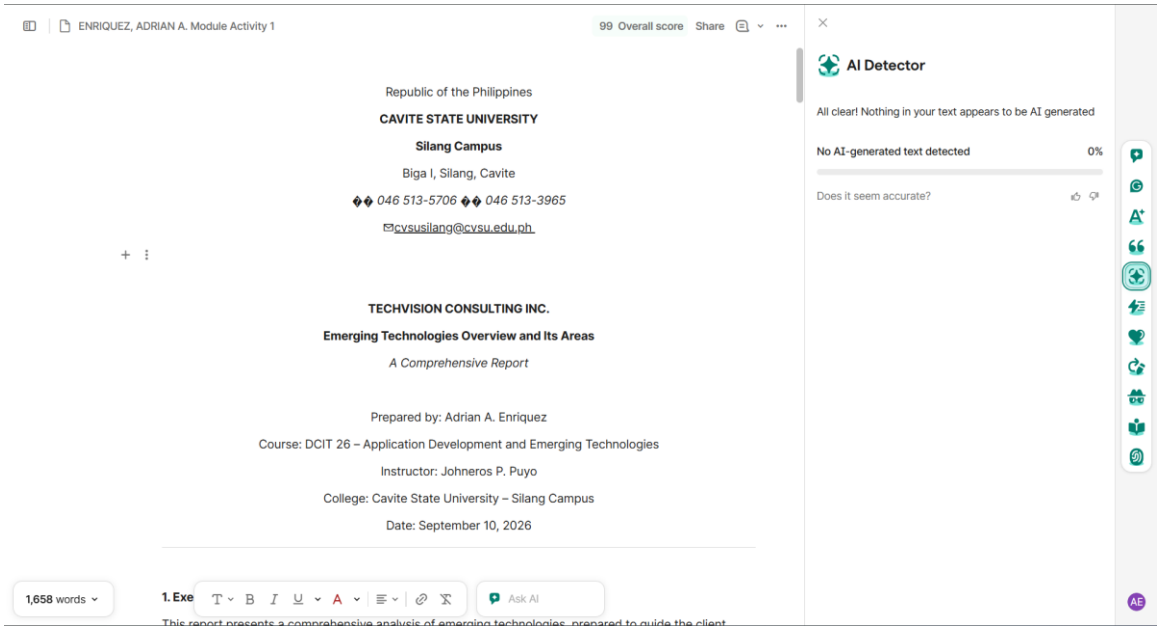
- Evaluate technologies against the five emergence characteristics. Before recommending adoption, evaluate a candidate technology's radical novelty, growth momentum, coherence with adjacent fields, potential impact, and residual uncertainties. Technologies that excel at the first four but remain ambiguous in their outcomes should get pilot projects rather than full-scale commitment.
- Consider convergence as a signal of opportunity. The coherence of several technologies such as the grouping of robotics, cloud, AI, and data analytics, indicates that integrated stacks are becoming more valuable than individual tools. The clients should design architecture that allows these components to work together.

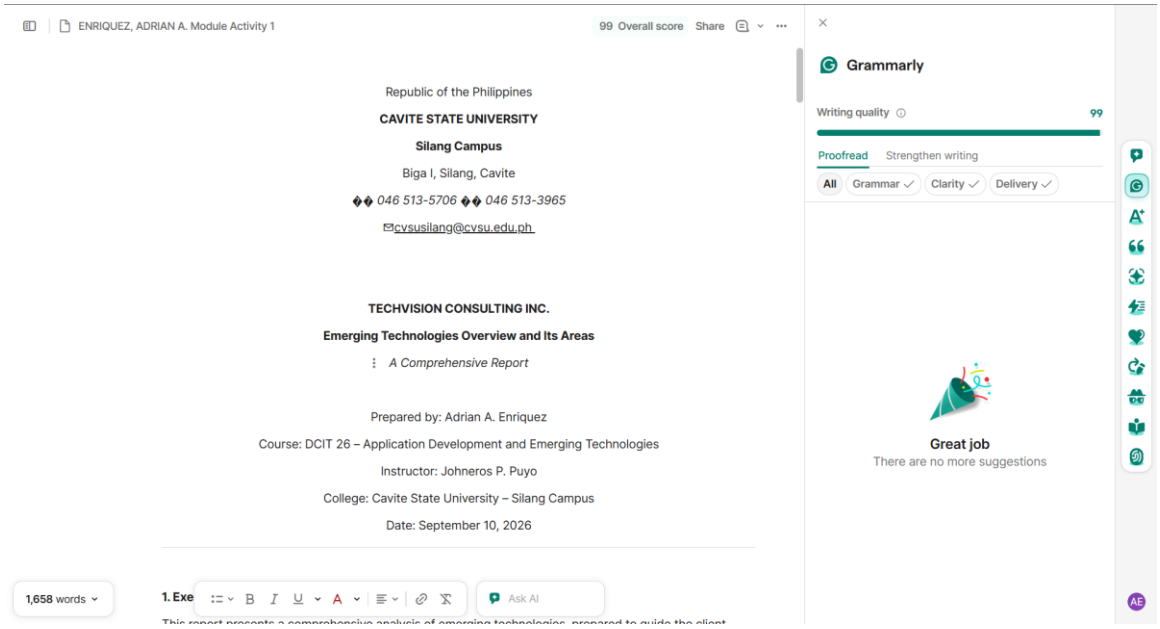
- Take advantage of Industry 4.0 building blocks to achieve quick operational gains. Cloud computing provides millisecond-reaction hosted services; IoT decentralizes analytics and decision-making for real-time response; and big data analytics enables quick answers from massive data for production and customer management.
- Keep an eye on quantum computing as a long-term bet. For most clients, quantum computing merits tracking rather than immediate investment due to its leading demonstrations from IBM, Google, Rigetti Computing, and IonQ, as well as its potential to help with materials design and simulate nature in the future.
- Every roadmap should take ambiguity and uncertainty into consideration. Client strategies should remain flexible since the significant effects of emerging technologies are still in the future and applications are still flexible, with potential unintended consequences.

8. Conclusion

Emerging technologies are best understood not just as new inventions but as radically novel and rapidly expanding fields, whether they are new or long-developing like artificial intelligence, that achieve coherence across previously distinct research streams and carry significant, system-wide impact that is still somewhat unclear and ambiguous. The technological landscape in which our clients will compete is defined by the industry 4.0 building blocks (big data analytics, IoT, cloud computing, and AR) and the areas highlighted by the World Economic Forum (AR, AI, machine learning, and quantum computing). As the emergence phase progresses, TechVision Consulting Inc. should advise clients to implement these technologies gradually, through pilot projects, and with ongoing evaluation.

9. Originality and Language Quality Assurance





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